

North American Logistics Operations (NALO)

Electrical Safety Awareness - Overview

Working around electricity and electrical equipment poses hazards for all employees.

In this course, learners will:

- **Be able to identify hazards and safe work practices in the work area**
- **Know which tasks they may and may not perform**

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Classifications of Electrical Workers

There are 2 classifications of electrical workers:

- Unqualified Affected Persons
- Qualified Electrical Workers (QEW).

The following slides will discuss the difference in tasks performed by each type of electrical worker.

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Unqualified Affected Person

An **Unqualified Affected Person** is any employee who works around electrical equipment but does not meet the qualifications to be a Qualified Electrical Worker.

Tasks / activities involving electrical equipment that an Unqualified Affected Person may perform are limited to those covered on the next slides.

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Tasks an Unqualified Affected Person MAY Perform

Operation of 120-volt power switches

Operation of Local start/stop switches

Operation of Emergency-stop buttons

Plug and unplug 120 volt and 480-volt receptacles, agitator cords, and battery charger cords

Access a control panel to operate disconnects.

NOTE: An Unqualified Affected Person MAY NOT remove faceplates or otherwise open control panels.

Reset (one time only) any reset button that is accessible without opening an electrical panel.

NOTE: Additional resets, after one reset, must be performed by a qualified electrical worker.

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Open junction boxes	Apply hard wire jumpers
Operate any circuit breaker	Reset Uninterruptible Power Supplies (UPS) overloads
Work on, or near exposed, energized parts	Replace blown fuses
Work on parts that are de-energized, but not locked out / tagged out	Work on load center gear in electrical switch rooms
Reset electrical equipment that requires a QEW after 1 attempt	Work on electrical circuits over 480-volts
Remove electrical panel guards	Reset, close, open or trip 480-volt load center feed breakers
Enter any electrical vaults / rooms	Establish temporary wiring
Perform electrical work on 24-, 120-, or 480-volt electrical equipment	Perform electrical maintenance and repair
Perform energized electrical work on a component of an electrical power system within reaching distance of exposed energized parts	Operation of 480-volt equipment disconnects
Open an electrical panel door (to view panel schedule)	

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Qualified Electrical Worker

Qualified Electrical Worker is an employee who meets the following qualifications:

- Has completed required QEW training requirements per established global and local curriculum (including documented supervisory approval as outlined in local training course)
- Has met at least one of the following criteria:
 - Four-year electrical engineering degree
 - Completion of a four-year apprenticeship
 - Two-year technical degree within an electrical discipline, plus related experience
 - Lilly shop / craft training and advancement through job classifications
 - Supervisor has certified that the employee is qualified based on experience and demonstrated ability to safely perform QEW tasks.

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Working Safely Around Electricity – General Guidelines

This section covers key guidelines to follow to decrease your chances of injury while working with the following:

- Electrical equipment
- Cords, cables and wires
- Plugs, extension cords and power strips

General guidelines to follow:

- Avoid wearing jewelry or metal objects when working with electricity
- Use safety features like three-pronged plugs, double-insulated tools and safety switches

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Working Safely Around Electricity – Guidelines for Electrical Equipment

Follow these guidelines for working safely with electrical equipment:

- Ensure that electrical equipment is maintained properly:
 - Regularly inspect tools, cords, grounds, and accessories
 - Check for loose, deformed, or missing parts
 - Inspect for damaged jackets or insulation
- Ensure that machine guards are in place
- Follow Lockout / Tagout procedures

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Working Safely Around Electricity – Guidelines for Electrical Equipment

Follow these guidelines for electrical equipment:

- Ensure damaged electrical equipment is removed from use.
- Do not modify electrical equipment from the manufacturers original design.
- Do not remove grounding prong from plugs.
- Do not modify or repair broken antennas with coat hangers, wires, etc. Remove broken antennas from service.
- Do not attempt to repair electrical equipment with tape, glue or other methods.

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Working Safely Around Electricity – Guidelines for Cords, Cables and Wires

Follow these guidelines for working safely with cords, cables and wires:

- Use only cords that are UL certified, and use the cord rated for the tool or equipment you are using.

Note: Cable rating information is printed on the cord. If the rating information is not legible, remove the cord from service.

- Do not staple or otherwise fasten a cord in a way that causes damage to its outer jacket.
- Keep electric cables and cords free from pinch points or kinks.
- Do not raise, lower, or carry equipment by its cord, or unplug equipment by pulling on its cord.
- Notify supervision or maintenance if signs of damage or deterioration are identified.

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Working Safely Around Electricity – Guidelines for Extension Cords

Follow these guidelines for working safely with extension cords:

- Do not use extension cords on a permanent basis. An extension cord may be used as temporary wiring (less than 30 days, per Lilly requirements) and if it complies with the National Electric Code.
- When portable tools are used with an extension cord, a GFCI must be used.
- Inspect for damage prior to use.

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Working Safely Around Electricity – Guidelines for Outlets and Surge Protectors / Power Strips

Follow these guidelines for working safely with outlets and surge protectors / power strips:

- Never overload any outlet. More than two plugs per double outlet is considered overloaded.
- Surge protectors / power strips can easily be overloaded and should be used only for low wattage equipment (such as a calculator, computers, desktop staplers, small fans, etc.). They must never be used for laser printers, appliances, copiers, or pedestal fans.

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Working Safely Around Electricity – Inspection of Electrical Equipment Prior to Use

Electrical Safety is an important element of a comprehensive fire safety program. In order to maintain a safe environment, an initial inspection is required for electrical equipment arriving at a Lilly site.

The purpose of the inspection is to:

- assure the equipment meets UL, FM (Factory Mutual) or equivalent standards
- has no apparent defects or broken / worn parts
- is properly grounded as required.

Equipment issued by Lilly will be inspected by Lilly QEW or a QEW agent of Lilly prior to issuance.

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Working Safely Around Electricity – Inspection of Electrical Equipment Prior to Use

Non-Lilly provided electrical equipment (e.g., items brought in by employees such as an ipod, pencil sharpener, etc.) no longer require an electrical inspection sticker.

But radios, fans and other heat generating electrical equipment still require an inspection.

Prior to bringing on site, employees ensure:

- The electrical equipment has been certified by a nationally recognized testing lab (ex. UL)
- The equipment is free of visual defects (if there is any question, contact maintenance supervision)

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Working Safely Around Electricity – Inspection of Electrical Equipment Prior to Use

Employee provided equipment must also be routinely inspected by the user for any damage to the cord or obvious hazards. If any damage or hazards are found, the device must be removed from service immediately.

Heat generating equipment may only be kept in designated break areas. Items as listed below are no longer permitted:

Portable heaters

Foot warmers

Coffee cup warmers

Electric blankets

Heating pads

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Working Safely Around Electricity – Inspection of Electrical Equipment Prior to Use

The following appliances are only permitted in “Approved break areas”:

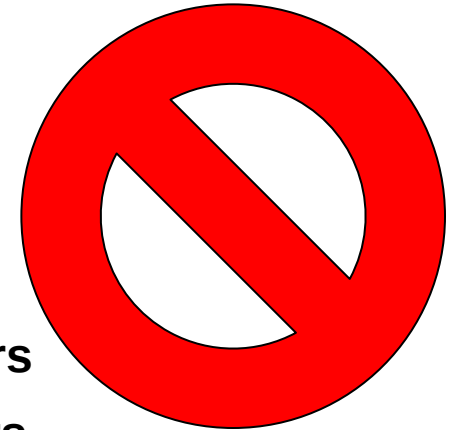
Coffee Pots
Microwaves
Toasters

Refrigerators
Toaster Ovens (special requirements)
Coffee Cup Warmers

Note: The maintenance and housekeeping of these units will continue to be monitored.

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Access to Electrical Panels



Never block access to electrical panels such as:

- Circuit breaker panels
- Switchboards
- Load centers
- VFD Cabinets
- Motor control centers
- Combination Starters
- Control panels

In many areas, there are signs, bollards and / or markings on the floor in front of electrical panels.

Equipment such as carts, pallets, skids, chairs, trash receptacles, or PITs must **never** be placed in front of electrical panels, even temporarily. A minimum of 36" of clearance should be allowed.

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Electrical Safety Awareness -Examples

Examples of situations involving electrical safety are covered on the next pages.

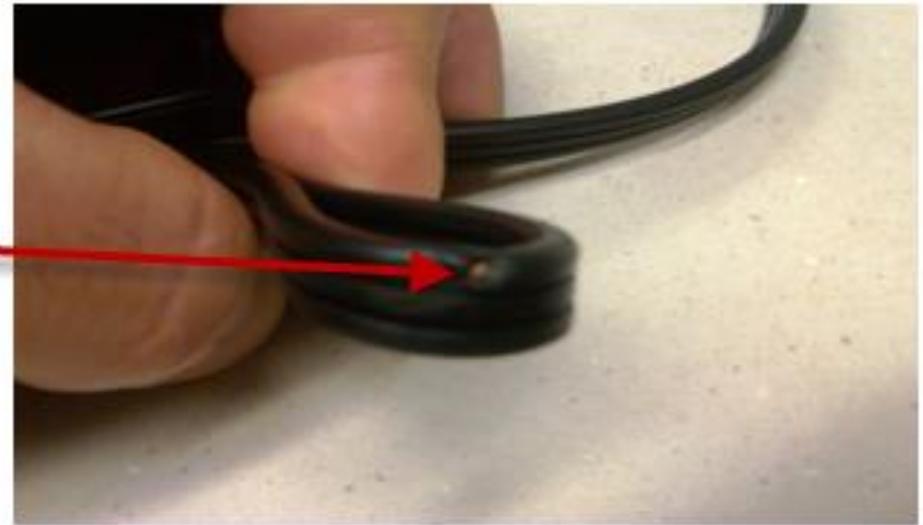
Employees should be aware of situations like these and report any suspected hazards to supervision or maintenance immediately.

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Electrical Safety Awareness - Examples

These pictures show cords that were not aligned properly in the cable tray, causing a pinch point and damage to a cord.

The situation had the potential for a shock from an electric short or arc.



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Electrical Safety Awareness - Examples

This picture shows a situation where access to electrical panels was blocked, even though signage exists.

This can be hazardous to personnel, equipment and facilities.



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Course Summary

To be safe around electrical equipment, every employee must:

- Perform ONLY the tasks that they are qualified to do
- Know the hazards and follow safe work practices
- Be aware of your environment and report any unusual situations or conditions

Please return to your learning plan and choose the “Agree” icon to receive credit for completing this course.

Document Approval Signatures

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